

Analysis of Used and New Car Dealership Data



Used Car Dealership | Data Analysis Report

Pricing • Inventory • Sales Strategy

1. Executive Summary

This report analyzes vehicle data from a dealership that sells both used and new cars. It examines the relationship between price and year of manufacture, the effect of mileage on price, and the distribution of fuel types across the inventory — providing insights that help management make better decisions on pricing, inventory, and sales strategy.

- **Price vs. year:** newer cars tend to be more expensive — a direct (positive) relationship between price and year of manufacture.
- **Price vs. mileage:** higher mileage generally leads to lower car prices — an inverse relationship.
- **Fuel type:** petrol is the most common fuel type (51.0%), followed by diesel (26.5%) and hybrid (22.5%).

2. Objectives and Research Questions

1. What is the relationship between car price and year of manufacture? Is it a direct relationship?
2. How does mileage affect the car price?
3. What is the distribution of fuel types among cars in the dealership?

3. Findings

3.1 Price vs. Year of Manufacture

The scatter plot shows a clear upward trend: as the year of manufacture increases, prices rise sharply. Cars built before 2000 mostly sell below 20,000, while models from 2015 onward frequently exceed 60,000, with recent models reaching 160,000. This confirms a direct relationship.

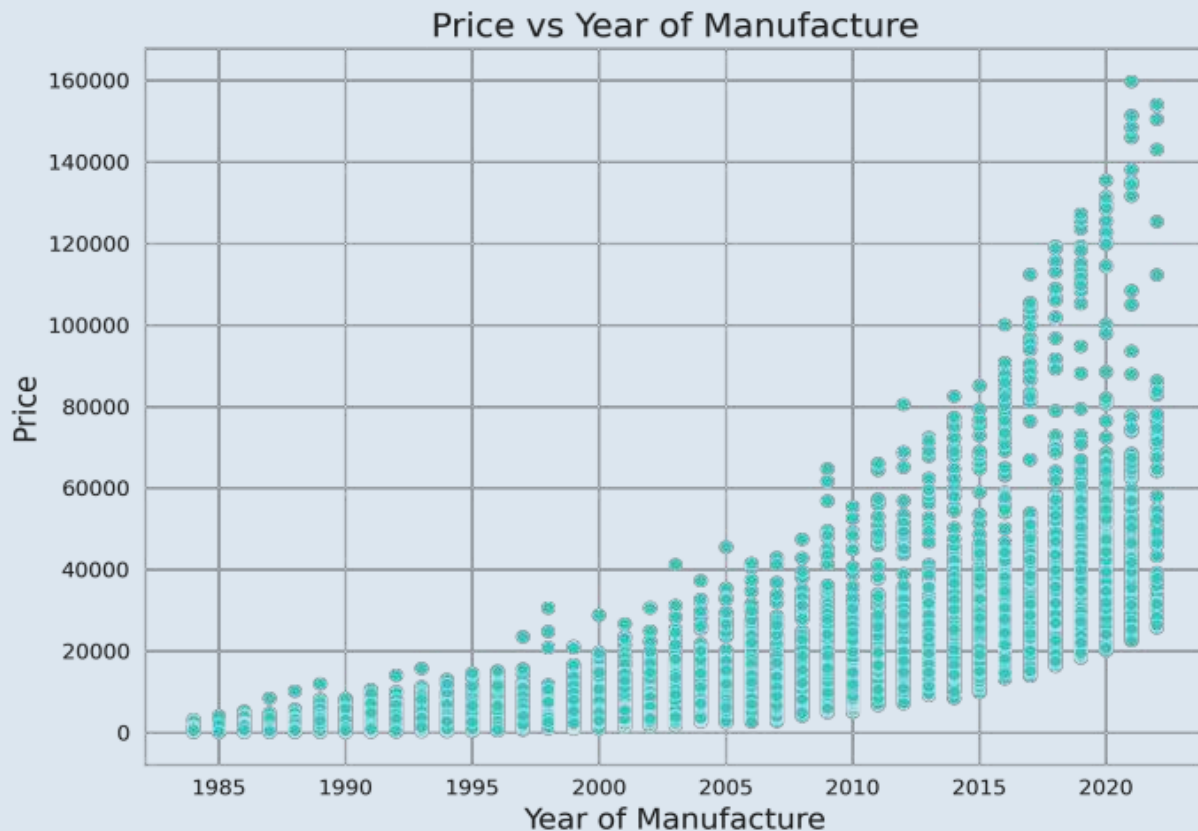


Figure 1 — Price vs. Year of Manufacture

3.2 Price vs. Mileage

The relationship between price and mileage is inverse. Low-mileage cars command the highest prices, and prices fall steeply as mileage increases. Beyond roughly 200,000 in mileage, nearly all vehicles cluster at the low end of the price range.

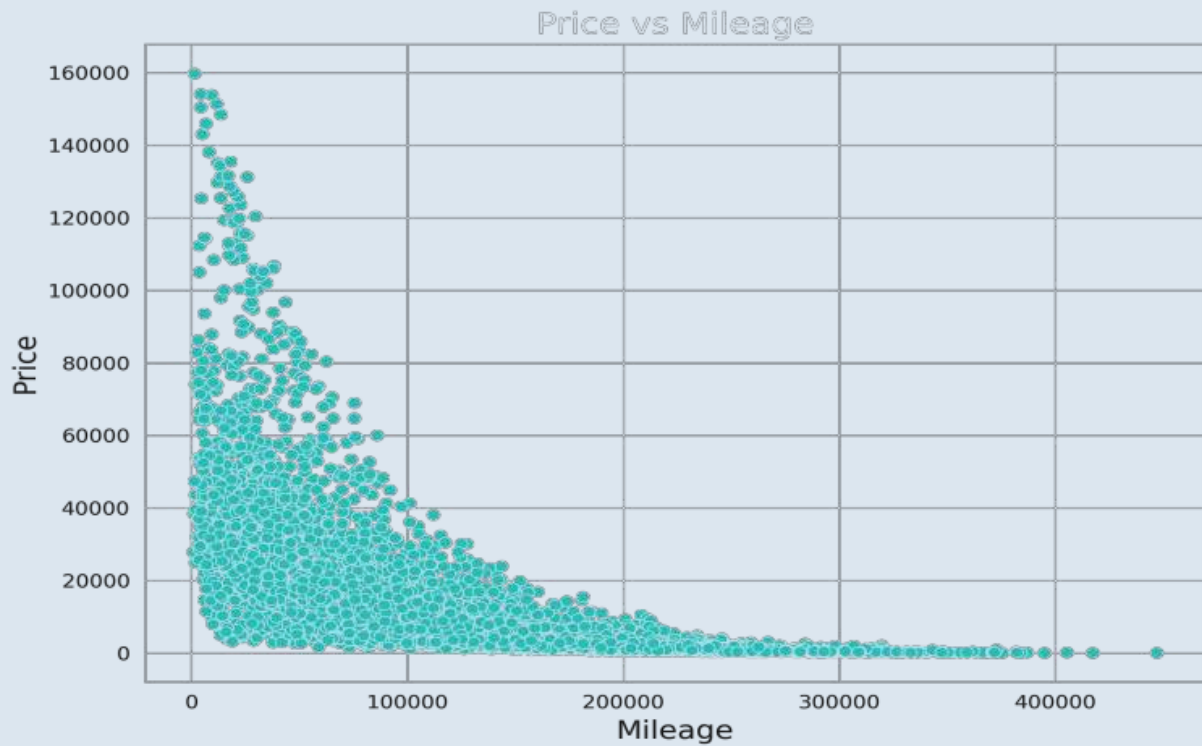
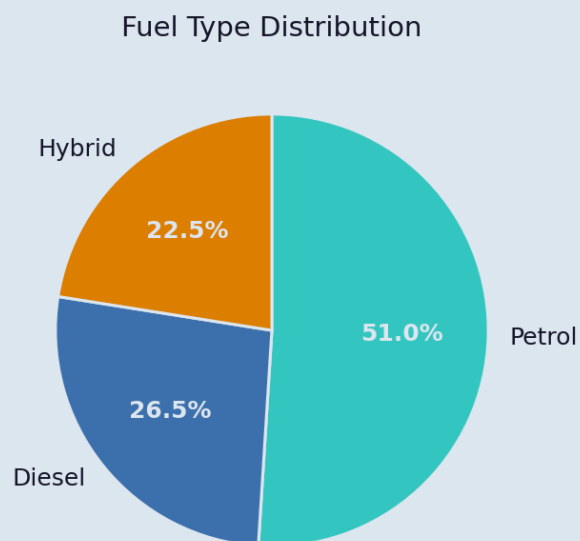


Figure 2 — Price vs. Mileage

3.3 Fuel Type Distribution

Petrol vehicles make up just over half of the inventory. Diesel accounts for about a quarter, and hybrids make up the remainder.



Fuel Type	Share
Petrol	51.0%
Diesel	26.5%
Hybrid	22.5%

4. Conclusions

1. Newer cars tend to be more expensive, confirming a direct relationship between price and year of manufacture.
2. Higher mileage generally leads to lower car prices.
3. Petrol is the most common fuel type among cars in the dealership.

5. Recommendations

1. **Pricing:** use year of manufacture and mileage as the two primary drivers when setting and reviewing vehicle prices.
2. **Inventory:** prioritize newer, low-mileage vehicles, which hold the highest value; discount high-mileage stock to move it faster.
3. **Sales strategy:** keep petrol models as the core of the stock mix while monitoring demand for hybrids as a growth segment.